Riddhi S. Patel

Cybersecurity Analyst | SOC Analyst | IT Support | CompTIA Security+ | Clearance Eligible | U.S. Citizen
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SUMMARY

Entry-level Cybersecurity Analyst with an M.S. in Cybersecurity Management and CompTIA Security+. Hands-on SOC experience in a large enterprise manufacturing environment, performing security monitoring, threat detection, SIEM-supported log analysis, vulnerability management, and incident response. Proven ability to identify IOCs, support risk mitigation decisions, and protect sensitive data across business-critical systems.

EDUCATION

M.S. Cybersecurity Management | University of Alabama in Huntsville *GPA: 3.70 | May 2025*
Relevant Coursework: Network Security, Digital Forensics, Incident Response, Risk Management, Cryptography, Cloud Security

B.S. Information Systems Cybersecurity Track | University of Alabama in Huntsville *GPA: 3.60 | Dec 2023*

WORK EXPERIENCE

IT Support Intern | Cyburity | Huntsville, AL *July 2026 – Oct 2026*

- Shadowed IT support specialists on client-facing ticket resolution, observing triage, prioritization, and escalation workflows in a managed services environment.
- Performed hardware inventory and diagnostics on Windows and Linux endpoints using CLI utilities (lscpu, dmidecode, lspci, lsblk) to document CPU, memory, storage, and peripheral specifications.
- Created bootable USB installation media and deployed Ubuntu Linux on existing Windows hardware, including BIOS/UEFI configuration and boot order adjustment
- Documented device specifications and configuration details to support asset tracking and hardware lifecycle decisions

Cybersecurity Analyst Intern | Mazda Toyota Manufacturing | Madison, AL *May 2025 – Aug 2025*

- Provided support for continuous security monitoring by reviewing log data on critical infrastructure and leveraging Zabbix to streamline incident detection and response processes in a Tier 1 SOC environment.
- Utilized Nmap for comprehensive network scanning and service enumeration during vulnerability management cycles, assigning risk priorities through CVSS to inform targeted mitigation across departments, with findings relayed via Power BI dashboards.
- Correlated network traffic analysis with threat intelligence to strengthen identification of malicious activity; coordinated findings with departments to support incident handling and remediation activities.
- Integrated automated risk assessment capabilities with Power Apps and Python in Docker to support effective monitoring and reporting of security metrics.
- Supported audit processes by maintaining standardized security briefings and policy materials, utilizing best practices in documentation and ensuring alignment with NIST and DoD RMF frameworks.

PROJECTS

SOC Home Lab | Security Monitoring & Incident Investigation | *Personal Cybersecurity Project |* *Sep 2026 - Present*

- Configured Sysmon and Windows event logging to capture process creation, PowerShell activity, network connections, and other security-relevant events.
- Investigated a simulated suspicious PowerShell activity scenario by correlating Sysmon Event ID 1 with PowerShell Event ID 4104 and reviewing process trees, command lines, user context, and timestamps.
- Conducted network reconnaissance using Nmap and Linux networking tools to identify listening services, investigate SMTP traffic on TCP port 25, & examine firewall activity.
- Analyzed Ubuntu UFW firewall logs to investigate repeated blocked inbound connection attempts and accessed service requirements & security risk before recommending firewall changes

Entropy Generation & Cryptographic System Validation | *Python, SHA-256, NIST SP 800-90B*

- Evaluated CPU jitter-based entropy sources for cryptographic use, applying NIST SP 800-90B assessment methodology — directly mirroring compliance validation work performed in security audits.
- Developed Python scripts to process entropy data and applied SHA-256 hashing to reduce bias; analyzed min-entropy across VMs, WSL containers, and bare-metal systems, producing documented security findings.

Security Monitoring & Network Traffic Analysis | *Zabbix, Wireshark*

- Simulated SOC Tier 1 workflows by performing alert triage on threshold breaches and behavioral anomalies in Zabbix, prioritizing events by severity and documenting escalation decisions.
- Captured and dissected live network traffic in Wireshark to identify suspicious protocol behavior and abnormal communication patterns; produced a structured incident investigation report.

LEADERSHIP & ACTIVITIES

NAC-ISSA (North Alabama Chapter — ISSA) *Active Member, 2025–Present*

- Networking with cybersecurity practitioners across North Alabama, attending industry events and building professional connections in the local security community.

CTF Competitions & TryHackMe

- Competed in CTF challenges (web exploitation, cryptography, forensics, network analysis) and actively completing TryHackMe labs in SOC analysis, threat hunting, and penetration testing.

CERTIFICATIONS

- **CompTIA Security+ (SY0-701)** • **AWS Certified Cloud Practitioner (2024)** • **AWS Certified Machine Learning Specialty (2025)**

TECHNICAL SKILLS

SIEM & Monitoring: Zabbix, Wireshark, Splunk (academic), SIEM, Windows Event Logs, BIOS/UEFI configuration
Threat & Vulnerability: Nmap, Alert triage, IOC identification, Threat Detection, CVSS, Vulnerability Scanning, Cross-Team Collaboration
Frameworks & Compliance: NIST CSF, NIST SP 800-90B, DoD RMF, Incident Response, Risk Assessment, Technical Documentation
Cloud & Scripting: AWS, Docker, Python, PowerShell, SQL, R, Hardware troubleshooting, OS installation and imaging